VI SEMESTER (OPEN ELECTIVE-II)	L	T	P	C
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20ME6O02 - FARM MACHINERY				

COURSE OBJECTIVES:
The objectives of this course are
1. To impart the students to understand the fundamentals of machinery in farming.
2. To enable the students to acquire knowledge on tillage and equipment used.
3. To introduce the students about various types of earth moving equipment.
4. To enable the students to acquire knowledge on seeding and spraying equipment.
5. To introduce the fundamentals of transplanting machinery and fertilizer equipment.

COURSE OUTCOMES:
After successful completion of this course, students should be able to:
CO1: Explain various types of machinery in farming. [K2]
CO2: Illustrate types of farm operation for craft cultivation with scientific understanding. [K2]
CO3: Explain various types of earth moving equipment. [K2]
CO4: Summarize various seeding methods and sprayer types. [K2]
CO5: Explain transplanting methods and fertilizer equipment. [K2]

UNIT I
FARM MECHANIZATION:
Farm mechanization- objectives of farm mechanization, sources of farm power, classification of farm machines. Materials of construction and heat treatment. principles of operation and selection of machines used for production of crops, Field capacities of different implements and their economics, Problems on field capacities and cost of cultivation

UNIT II
TILLAGE EQUIPMENT:
Tillage equipment - classification and types of tillage, Primary tillage implements-mould board plough and its parts, disc plough, and other ploughs, Secondary tillage equipment- disc harrows, Implements-cultivators, intercultural implements. Forces acting on tillage tools, Problems on forces analysis, Draft measurement of tillage equipment, Draft and unit draft related problems.

UNIT III
EARTH MOVING EQUIPMENT
Earth moving equipment - terminology, construction and their working principles, shovels, bulldozers, trenches and elevators.

UNIT IV
SEED DRILLS AND SPRAYER:
Seeding - methods, types of seed metering mechanism, types of furrow openers. Calibration of seed drills, Adjustment of seed drills – objectives, uses of plant protection equipment
Sprayers - types of sprayers and dusters, sprayer calibration and selection, Constructional features of different components of sprayers and dusters

UNIT V
TRANSPLANTING AND FERTILIZER:
Transplanting and fertilizer - transplanting methods, different types of transplanting machinery, working principle, adjustments in transplanting equipment
Fertilizer - application equipment, fertilizer meeting mechanism calibration of fertilizer equipment.

Text Books:

1. Fakir Chara Das, Kishore Chandra and Shishira Kanth, Farm Machinery and Equipment, 1st Edition, Akinik Publications, 2020.
2. Triveni Prasad Singh, Farm Machinery, 1st Edition, Prentice Hall India Pvt, Limited, 2016.

Reference Books:

1. Surendra Singh, Farm Machinery Principal And Applications, 1st Edition, ndian Council of Agricultural Research, 2017.
2. Smith H P, Farm Machinery and Equipment, Tata McGraw Hill Publishing Co. Ltd., New Delhi, 2007.

Web Reference:

1. https://nptel.ac.in/courses/126105009